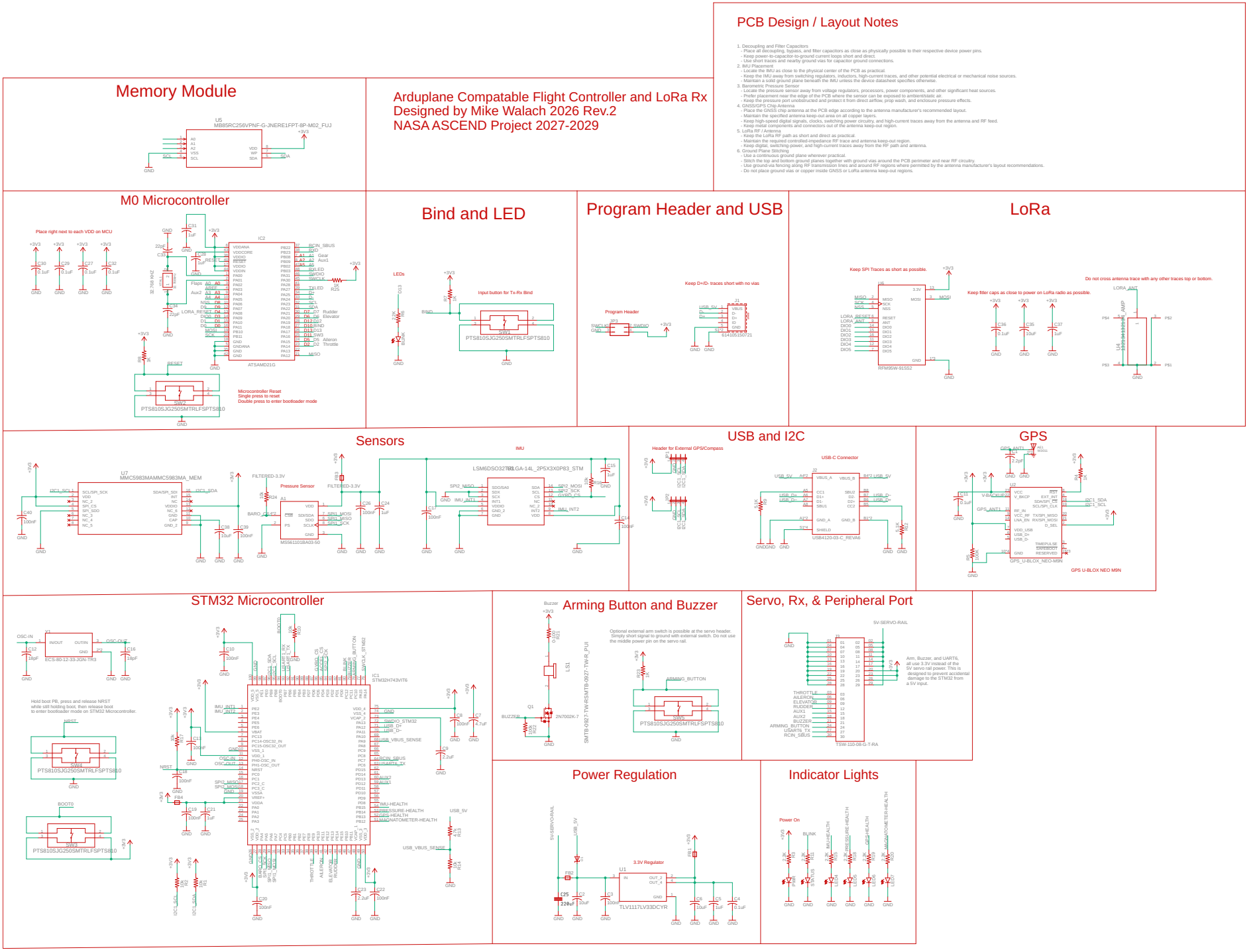

PCB Design / Layout Notes
1. Decoupling and Filter Capacitors
- Place all decoupling, bypass, and filter capacitors as close as physically possible to their respective device power pins.
- Keep power-to-capacitor-to-ground current loops short and direct.
- Use short traces and nearby ground vias for capacitor ground connections.
2. IMU Placement
- Locate the IMU as close to the physical center of the PCB as practical.
- Keep the IMU away from switching regulators, inductors, high-current traces, and other potential electrical or mechanical noise sources.
- Maintain a solid ground plane beneath the IMU unless the device datasheet specifies otherwise.
3. Barometric Pressure Sensor
- Locate the pressure sensor away from voltage regulators, processors, power components, and other significant heat sources.
- Prefer placement near the edge of the PCB where the sensor can be exposed to ambient/static air.
- Keep the pressure port unobstructed and protect it from direct airflow, prop wash, and enclosure pressure effects.
4. GNSS/GPS Chip Antenna
- Place the GNSS chip antenna at the PCB edge according to the antenna manufacturer's recommended layout.
- Maintain the specified antenna keep-out area on all copper layers.
- Keep high-speed digital signals, clocks, switching power circuitry, and high-current traces away from the antenna and RF feed.
- Keep metal components and connectors out of the antenna keep-out region.
5. LoRa RF / Antenna
- Keep the LoRa RF path as short and direct as practical.
- Maintain the required controlled-impedance RF trace and antenna keep-out region.
- Keep digital, switching-power, and high-current traces away from the RF path and antenna.
6. Ground Plane Stitching
- Use a continuous ground plane wherever practical.
- Stitch the top and bottom ground planes together with ground vias around the PCB perimeter and near RF circuitry.
- Use ground-via fencing along RF transmission lines and around RF regions where permitted by the antenna manufacturer's layout recommendations.
- Do not place ground vias or copper inside GNSS or LoRa antenna keep-out regions.
Memory Module
U5
MB85RC256VPNF-G-JNERE1FPT-8P-M02_FUJ
Arduplane Compatable Flight Controller and LoRa Rx
Designed by Mike Walach 2026 Rev.2
NASA ASCEND Project 2027-2029
M0 Microcontroller
Place right next to each VDD on MCU
ATSAMD21G
Microcontroller Reset
Single press to reset
Double press to enter bootloader mode
Bind and LED
LEDs
Input button for Tx-Rx Bind
Program Header and USB
Program Header
Keep D+/D- traces short with no vias
LoRa
Keep SPI Traces as short as possible.
Keep filter caps as close to power on LoRa radio as possible.
Do not cross antenna trace with any other traces top or bottom.
RFM95W-915S2
Sensors
IMU
MMC5983MAMMC5983MA_MEM
Pressure Sensor
LSM6DSO32TR
MS5611018A03-50
USB and I2C
Header for External GPS/Compass
USB-C Connector
USB4120-03-C_REVA6
GPS
GPS_U-BLOX_NEO-M9N
GPS U-BLOX NEO M9N
STM32 Microcontroller
STM32H743VITx
Hold boot PB, press and release NRST
while still holding boot, then release boot
to enter bootloader mode on STM32 Microcontroller.
Arming Button and Buzzer
Optional external arm switch is possible at the servo header.
Simply short signal to ground with external switch. Do not use
the middle power pin on the servo rail.
Servo, Rx, & Peripheral Port
5V-SERVO-RAIL
Arm, Buzzer, and UART6,
all use 3.3V instead of the
5V servo rail power. This is
designed to prevent accidental
damage to the STM32 from
a 5V input.
Power Regulation
3.3V Regulator
TLV1117LV33DCYR
Indicator Lights
Power On
BLINK